Resolution Geometry: The Impossibility of the Perfect Window

Why Better Instruments Do Not Remove the Frame, and Why the Frame Needs Its Own Mathematics

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Abstract

This paper has two parts and two audiences.

Part I. Ten short scenes explain, without mathematics, what it means for an observer to see the world through a window.

Part II. The rest of the paper is written for the scientist who decides to check. Each scene becomes a layer with three elements:

- a formal statement with proof;
- internal critique loops, each consisting of an objection, a reply and the correction it forced;
- the established tool that fails silently there, and the RG object that does not.

Assembly. The layers are assembled into a scoped *No-Perfect-Window theorem*. Refinement of instruments never by itself establishes observational completeness. Six independent, checkable conditions each force a nontrivial fibre modulo gauge:

(i) a symmetry that the accessible protocols cannot break;
(ii) finite record capacity;
(iii) a causal horizon;
(iv) global classes probed by local patches;
(v) reduced access to a subsystem;
(vi) finite precision under chaotic development.

In our universe the measured $\Lambda > 0$ (Pantheon+, $\Lambda = 0$ rejected at $\Delta\chi^2 = 48$) gives every comoving observer a finite event horizon, of comoving radius 16.2 Gly today in flat ΛCDM. Under that extrapolation, condition (iii) holds for every comoving observer.

External critique. The assembled argument is exposed to eight disciplines: statistics, experimental physics, relativity, quantum information, topology, philosophy, computation, and the novelty skeptic. Each critique forces a logged correction.

Result. What survives is a converted burden of proof. A claim that a perfect physical observer is possible is a theorem to be proved: it requires the observation map and a proof that its fibres are trivial modulo gauge. RG is the bookkeeping that such a proof, or its failure, needs.

The world may be complete. An observer's access to it need not be.

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Part I
The window, for everyone

1 Ten scenes

1. The keyhole. Through a keyhole you see a red ball. A better lens shows its scratches, but not what stands behind the cupboard. *Seeing more detail and removing the edge of sight are different achievements.*

2. Two rooms that look the same. In one room a hidden wheel turns clockwise, in the other anticlockwise. Every reading you can take is identical. No amount of cleverness applied to identical readings can tell the rooms apart. *Computation cannot recover what never entered the record.*

3. The better instrument. You open the panel and add a sensor. Now the rooms differ, and that is progress. But the new sensor has its own place, noise and channel. *Improving a window changes the window. It does not abolish windows.*

4. The observer inside. To watch the whole house you must stand somewhere. Outside the house, you are not part of what you describe. Inside, you are one more object with a limited view. *Information can be preserved in the whole and still be out of the observer's reach.*

5. Two questions, one measurement. Asking "where" and "how fast" of a delicate object can require different handling, and answering one sharply can blur the other. *Which question is asked shapes what can be learned.*

6. The notebook. A notebook with a hundred pages cannot record a million different answers. *A finite recorder has finite room.*

7. The horizon. Some places are so far, in a universe that is speeding up, that no signal from them will ever arrive here. A bigger telescope does not help, because nothing is coming. *Some darkness is not behind an object. It lies outside the reach of any signal.*

8. The building of rooms. Everyone knows their own room perfectly, and every doorway lines up. Still, no one inside a single room knows the shape of the building. A corridor might loop back with a twist. *Perfect local knowledge is not global knowledge.*

9. Tomorrow. Knowing today very well is not knowing next year. A tiny blur can grow until it decides the outcome. *Precision about the present is not possession of the future.*

10. Four kinds of dark. Some darkness is only a different name for the same thing. Some is a shadow that a second lamp removes. Some is the shape of the whole building. Some is a door that cannot be built at all. *They are different, and they need different remedies.*

The world may be complete. An observer's access to it need not be.
The right question is not whether science will eventually see everything.
It is what a declared observation determines, what remains compatible with it,
and which further observation would tell the alternatives apart.

Part II
The argument, layer by layer

2 Setting

Definition 2.1 (Protocols, families, fibres, physical equivalence)**.** The basic objects are these.

- A *protocol* p maps physical possibilities $x \in X$ (preparations or histories) to laws of records, $\mathcal{O}_p : X \to \mathcal{P}(R_p)$.
- For a family $\mathscr{P}$, write $x \sim_{\mathscr{P}} y$ iff $\mathcal{O}_p(x) = \mathcal{O}_p(y)$ for all $p \in \mathscr{P}$. The classes are the *fibres* of $\mathscr{P}$.
- *Physical equivalence* $\sim_{\text{phys}}$ identifies representations that the declared theory treats as the same: gauge, coordinates, isometry.
- A *task* is a function Q on $X/\!\sim_{\text{phys}}$.
- Q is *determined* by $\mathscr{P}$ iff it is constant on the fibres of $\mathscr{P}$.

Definition 2.2 (The perfect-window hypothesis)**.** An observer with admissible family $\mathscr{P}$ has a *perfect window* if

$$x \not\sim_{\text{phys}} y \implies \exists p \in \mathscr{P} :\ \mathcal{O}_p(x) \neq \mathcal{O}_p(y). \tag{PW}$$

Everything below is about whether (PW) can be *inferred*, and which physically checkable conditions *refute* it for a specified observer.

3 Layer 1: identical records

Proposition 3.1 (Two-world bound)**.** *If $\mathcal{O}_p(A) = \mathcal{O}_p(B)$, then every decision rule δ applied to the records of p satisfies*

$$\max\big(\mathbb{E}_A|\delta - Q(A)|,\ \mathbb{E}_B|\delta - Q(B)|\big)\ \geq\ \tfrac{1}{2}|Q(A) - Q(B)|.$$

Proof. δ has the same law in both worlds. By the triangle inequality, $\mathbb{E}|\delta - Q(A)| + \mathbb{E}|\delta - Q(B)| \geq |Q(A) - Q(B)|$. Equality is attained at the midpoint (W1). □

Objection. "This is trivial."
Reply. It is trivial, and that is its function. It turns "we will know eventually" into a statement that must be proved: that the relevant fibres are trivial.
Correction applied. The proposition is presented as the *definition of the burden*, not as a discovery.

Objection. "A Bayesian prior picks one world."
Reply. A prior is not a record. With identical records, the posterior restricted to the fibre equals the prior restricted to the fibre. The answer is then a declaration, not an observation.
Correction applied. RG types prior-selected answers as *declared*, never as *determined*.

Silent failure. Point estimation without an identifiability check returns a number on a nontrivial fibre. Examples are the alias at $3M$ samples and Fisher-null directions that are not null (Engagement S4, S7). *RG object:* the fibre, together with typed abstention.

4 Layer 2: refinement is not completeness

Proposition 4.1 (Refinement shrinks kernels; it need not reach zero)**.** *For smooth protocols,*

$$\ker D\mathcal{O}_{(p_1,p_2)} = \ker D\mathcal{O}_{p_1} \cap \ker D\mathcal{O}_{p_2},$$

and the joint fibres are intersections of fibres. A nested sequence of refinements may stabilize at a nontrivial fibre.

Theorem 4.2 (Symmetry-closed families never separate orbits)**.** *Let a group G act on X, and suppose every $p \in \mathscr{P}$ is G-invariant: $\mathcal{O}_p(gx) = \mathcal{O}_p(x)$. Then*

(a) every fibre of $\mathscr{P}$, and of every refinement within G-invariant protocols, contains whole G-orbits;

(b) if G moves some state to a physically distinct one, (PW) fails;

(c) separating the orbit requires a protocol outside the class, i.e. a new resource such as a shared reference frame.

Proof. (a) is immediate from invariance, and (b) and (c) follow. Witness (W2): without a shared reference frame, every qubit protocol factors through the twirl, which sends every state to $I/2$. This is exact. The general theory of reference frames and superselection is imported [1]. □

Objection. "Then share a reference frame."
Reply. Yes. That adds a physical resource, which is a change of declaration. The theorem forbids only the inference "refine within the class, therefore complete".
Correction applied. Resources that split fibres are logged as *declared resources* (RG's extension gate).

Objection. "In practice G is trivial."
Reply. Then the triviality must be shown, and that is the burden of Layer 1.
Correction applied. (PW) is restated as an obligation: exhibit $\mathscr{P}$ and prove triviality modulo gauge.

Silent failure. "More data of the same kind" is assumed to converge to the truth. *RG object:* the family quotient and the extension gate. Extension completeness is equivalent to local tomography (Rigidity).

5 Layer 3: the observer inside

Proposition 5.1 (Preservation in the whole is not access in the part)**.** *There are unitary joint evolutions whose reduced channel is not injective. For example, complete dephasing sends $|+\rangle$ and $|-\rangle$ to the same $I/2$ (Engagement S8).*

Imported Theorem 5.2 (Self-measurement [3, 4])**.** *An apparatus that is part of the system it measures cannot distinguish all states of that system, itself included. This is stated in the cited works under their stated hypotheses.*

Objection. "Measure the environment too."
Reply. Then the observer's domain has been enlarged, and the obligation moves to the new domain. Including the entire universe meets the self-measurement results.
Correction applied. The claim is restricted to *a declared observer domain.* No claim is made about an observer outside physics.

Silent failure. "The total evolution is unitary, so the information is recoverable." *RG object:* joint reversal versus reduced recovery, with Petz sufficiency as the exact criterion.

6 Layer 4: the question shapes the answer

Imported Theorem 6.1 (Joint measurability [2]). *Unbiased joint approximations of σ_x and σ_z with sharpness η_x and η_z exist iff $\eta_x^2+\eta_z^2 \leq 1$. W6 checks the boundary exactly: the minimum eigenvalue is 0 at $\eta = 1/\sqrt{2}$ and negative at $\eta = 0.75$.*

The quantum metric for a task is a band, SLD $\leq$ WY $\leq$ BKM $\leq$ RLD, and the task declares the member (Quantum Mechanics print).

Objection. "Measure sequentially."
Reply. The first measurement disturbs the second. The trade-off relations are imported.
Correction applied. Commuting tasks are explicitly exempt. Only incompatible task pairs carry the context cost.

Silent failure. A single "information" number is quoted for a quantum state without a declared task. *RG object:* the task lock.

7 Layer 5: the finite recorder

Proposition 7.1 (Capacity). *Records of k bits separate at most 2^k hypotheses. For probabilistic records, Fano's inequality gives an error of at least $1-(k+1)/\log_2 M$ among M hypotheses. W3 gives 0.45 for $M = 2^{20}$ and $k = 10$.*

Objection. "Bekenstein bounds are not proved in general."
Reply. Correct. The argument does not need them: every actual observer has a declared finite memory. Entropy bounds would make finiteness universal for observers in bounded regions.
Correction applied. The capacity condition is stated for *declared finite capacity.* The Bekenstein bound is cited only as a conditional strengthening [5].

RG object: capacity as part of the protocol declaration.

8 Layer 6: the causal horizon

Lemma 8.1 (Causal restriction). *Assume relativistic causality, so that the records of an observer segment O are functions of the fields on $J^-(O)$. If $W_1|_{J^-(O)} = W_2|_{J^-(O)}$, then $\mathcal{O}_O(W_1) = \mathcal{O}_O(W_2)$.*

Corollary 8.2 (Our universe has a permanent window). *In flat ΛCDM with $\Omega_m = 0.3326$, our Pantheon+ fit with $\Lambda = 0$ rejected at $\Delta\chi^2 = 48$, and $H_0 = 70$, every comoving observer has a finite comoving event horizon of 4970 Mpc, i.e. 16.2 Gly (W4). Events beyond it never enter any record kept on that worldline. Histories differing only there lie in one fibre forever.*

Objection. "This extrapolates Λ into the future."
Reply. Yes. The corollary is conditional on constant Λ, which is the measured model. Phantom or decaying dark energy changes the horizon, not the logic.
Correction applied. Corollary 8.2 is stated conditionally.

Objection. "Another observer can see beyond."
Reply. Each horizon is observer-relative. Records cannot be sent across it, so no single record system collects both sides.
Correction applied. The claim is "for every comoving observer", not "for no observer anywhere".

Silent failure. "A better telescope will show it." *RG object:* the permanent fibre, which RG types as a pool fibre in its spacetime papers.

9 Layer 7: local is not global

Proposition 9.1 (Locally identical, globally distinct)**.** *A cylinder and a Möbius band are covered by identical flat patches. The orientation holonomy around the core loop is* $+1$ *for the cylinder and* -1 *for the band (W5). More generally, local charts and overlap maps determine a Čech class. The global object exists iff the class vanishes, and protocols confined to patches cannot read the class.*

Objection. "Then walk around the loop."
Reply. A loop protocol is a global protocol, and RG types it as one. With finite samples it needs a declared coverage: the exact threshold is $\max(\pi/N, \pi/2 - 2\alpha)$, and below it aliases exist for every estimator (Spectral Identifiability).
Correction applied. The claim is that *patch-local protocols* do not determine global classes. Global protocols can, under declared coverage.

Silent failure. Local consistency is assumed to imply a global reconstruction. *RG objects:* rooms, obstructions, the nerve theorem and the coverage threshold.

10 Layer 8: development

Proposition 10.1 (Prediction horizon)**.** *With a positive Lyapunov exponent* λ *and initial precision* δ*, trajectories separate to order one after about* $\lambda^{-1}\log(1/\delta)$ *steps. W7: for the logistic map at* $r = 4$ *with* $\delta = 10^{-30}$*, separation occurs at step* 98*, against* $\log_2 10^{30} = 99.7$*. By Layer 5,* δ *is bounded below by the declared capacity.*

Classification does not select dynamics (L128): the same present-state invariant is compatible with many laws.

Objection. "Deterministic laws suffice in principle."
Reply. With infinite precision, yes. Physical records are finite. In addition, an invertible law alone forgets nothing (Rooting Closure), so the horizon belongs to the protocol, not to the law.
Correction applied. The prediction horizon is typed as a property of the *protocol*, never of the dynamics.

11 Layer 9: representation is not ignorance

Proposition 11.1 (Null directions)**.** *Coordinate or gauge differences are not physical distinctions. A metric component at a coordinate point varies across gauges while every relational prediction does not (Engagement S9; GR Without Spacetime).*

Objection. "Then the darkness is not real."
Reply. Correct for null directions. They are not ignorance, and RG types them separately.
Correction applied. (PW) is defined modulo $\sim_{\text{phys}}$, so that gauge never counts as blindness.

12 Layer 10: four kinds of darkness

Theorem 12.1 (The kinds are distinct, and conflation fails)**.** *Null (gauge), hidden (visible in histories or extensions), room (a global class not visible locally) and obstruction (a global extension class) are pairwise distinct. Each separation has a witness, given in the Engagement Case. A single undifferentiated notion of "measurement error" conflates at least two of them, and the matching silent failure from S1–S12 occurs.*

Remedies differ:

- null needs no remedy;
- hidden needs a richer history or extension;
- a room needs a global or joint protocol;
- an obstruction may admit no global object at all.

Part III
Assembly

13 The No-Perfect-Window theorem

Theorem 13.1 (No perfect window from refinement; sufficient obstacles)**.** *Let an observer be declared by four items: its causal domain D; its record capacity k; an admissible protocol family $\mathscr{P}$; and the patch structure of its protocols.*

(I) No inference from refinement within $\mathscr{P}$ establishes (PW) unless the joint fibres of $\mathscr{P}$ are proved trivial modulo $\sim_{\mathrm{phys}}$.

(II) Each of the following implies a nontrivial fibre modulo gauge, so (PW) fails:

- *(a) a symmetry G that leaves every protocol invariant and moves some state to a physically distinct one (Theorem 4.2);*
- *(b) more physically distinct classes than 2^k, or than allowed by Fano for a target error;*
- *(c) physically distinct histories agreeing on $J^-(D)$;*
- *(d) a nontrivial global class, with patch-confined protocols;*
- *(e) access only through a reduced channel that is not injective;*
- *(f) for prediction tasks: a positive Lyapunov exponent with precision bounded by k, beyond the horizon $\lambda^{-1}\log(1/\delta)$.*

(III) In flat ΛCDM with the measured $\Lambda > 0$, condition (c) holds for every comoving observer (Corollary 8.2).

Proof. (I) is Proposition 3.1 applied to each fibre. (II) collects Layers 2, 5, 6, 7, 3 and 8 in order. (III) is Corollary 8.2. □

Corollary 13.2 (The converted burden)**.** *A claim that a physical observer has a perfect window is a theorem with an obligation. It must supply the observation map of its family, and prove that the fibres are trivial modulo gauge on the declared domain, avoiding each of (a)–(f). Writing that obligation down, and checking it, is what RG is for.*

Part IV
External critique, branch by branch

14 Eight critics

The statistician. Objection. "This is identifiability, known for a century."
Reply. The core is identifiability. Classical identifiability is, however, per model and local. It does not type rooms (S1: equal Fisher fields, non-equivalent experiments; S5: zero local

loss, no global recovery) or obstructions, and it gives no exact abstention thresholds (S7, the $\max(\pi/N, \pi/2 - 2\alpha)$ threshold).
Correction applied. Correction C1: credit identifiability explicitly as the core. RG's added content is the cross-level typing and the sharp thresholds.

The experimentalist. Objection. "We always add detectors; this is pessimism."
Reply. Adding detectors is how hidden values become visible, and RG welcomes it. The theorem targets only the inference from refinement to completeness, and names three obstacles that refinement within a class cannot cross: symmetry (W2), capacity (W3) and horizon (W4).
Correction applied. Correction C2: hidden values are explicitly typed as *removable*. Layer 2 states that a new resource is a change of declaration.

The relativist. Objection. "Horizons and relational observables are textbook."
Reply. Agreed, and RG imports them. What is added is the typed statement that a horizon fibre is permanent for an observer while gauge is null, with the measured-Λ corollary attached to data.
Correction applied. Correction C3: Corollary 8.2 is made conditional on constant Λ and observer-relative.

The quantum information theorist. Objection. "Reference frames, superselection, Holevo and joint measurability are all known."
Reply. Agreed. They are imported, from [1, 2] among others. RG's addition is one ledger in which they sit beside horizon, topology and capacity obstacles, with the same fibre semantics.
Correction applied. Correction C4: the imports are cited at every use, and no quantum result is claimed as new.

The topologist. Objection. "Holonomy and Čech classes are classical."
Reply. Yes. The operational claim is that patch-confined protocols cannot read them, and that loop protocols need declared coverage, with a sharp threshold.
Correction applied. Correction C5: no new topology is claimed; the claim is about which protocols determine which classes.

The philosopher. Objection. "This is verificationism or anti-realism."
Reply. No. Meaning is not tied to verification, and the world may contain what no record captures. The thesis restricts *claims*, not *existence*.
Correction applied. Correction C6: the paper states explicitly that RG is compatible with realism ("the world may be complete") and rejects the verification principle.

The theorist of computation. Objection. "Is this Gödel or Turing in disguise?"
Reply. No logical incompleteness is claimed. Self-measurement limits are imported in their physical form [3, 4], under their hypotheses.
Correction applied. Correction C7: no diagonal argument is claimed; the self-inclusion layer is restricted to the declared domain.

The novelty skeptic. **Objection.** "Every ingredient is known."
Reply. Most are, and they are cited. The claim is that no single framework, to the author's knowledge, carries all six obstacles, the four kinds of darkness, the exact thresholds, and the forced local geometry (Čencov from descent) as one typed calculus. Naming one refutes the claim.
Correction applied. Correction C8: "unique" is replaced by "minimal typed calculus covering all ten layers jointly", offered for refutation.

15 Corrections log

	Before critique	After
C1	"RG discovers non-identifiability"	identifiability credited as the core; RG adds cross-level typing and thresholds
C2	"instruments cannot reveal hidden structure"	hidden is removable; only within-class refinement is limited
C3	"some events are unknowable"	conditional on constant Λ, and observer-relative
C4	quantum limits presented as RG results	imported and cited
C5	"local data never give global structure"	patch-confined protocols do not; global protocols with declared coverage can
C6	a tone suggesting anti-realism	explicitly realist-compatible; the verification principle rejected
C7	self-inclusion as a logical impossibility	physical self-measurement results, imported and scoped
C8	"unique machinery"	minimal joint typed calculus, offered for refutation

Part V
Why the frame needs its own mathematics

16 Door by door: old tool, silent failure, RG object

Door	Old tool	Silent failure	RG object	Why minimal
Identical records	point estimate	answers on a nontrivial fibre (S4, S7)	fibre, abstention	Prop. 3.1
Refinement	more data of the same kind	cannot cross a symmetry orbit (W2)	family quotient, extension gate	Thm. 4.2
Observer inside	"unitary, so recoverable"	reduced channel not injective (S8)	joint vs reduced recovery, Petz	Petz equality criterion
Context	one information number	task-dependent metric band	task lock	Petz band
Recorder	unlimited memory assumed	pigeonhole, Fano (W3)	capacity in the declaration	Fano
Horizon	"bigger telescope"	no signal ever arrives (W4)	permanent fibre	causal lemma

Door	Old tool	Silent failure	RG object	Why minimal
Building	local consistency implies global	cylinder vs Möbius (W5)	rooms, obstructions, coverage	nerve, Čech
Tomorrow	precise present implies future	horizon $\lambda^{-1}\log(1/\delta)$ (W7)	protocol-typed horizon	Rooting Closure
Gauge (repre-sentation)	coordinates as physics	g_{tt} gauge-dependent (S9)	null type	hole argument
Kinds of dark	one "error" notion	S1–S12	four-kind calculus	pairwise witnesses

Why RG and not a patchwork. Each door, taken alone, can be handled by a specialist tool: identifiability, reference frames, Fano, horizons, Čech cohomology, Lyapunov analysis, gauge theory. But the doors compose. A real experiment meets several of them at once. Composition needs one semantics, "constant on the fibres of a declared family", in which the obstacles add as intersections of fibres. RG supplies that semantics, together with its forced local geometry (Čencov from descent) and its sharp thresholds. A patchwork has no common notion of fibre, so it cannot add obstacles, and its silent failures occur exactly at the seams.

17 What this paper does not claim

- That better instruments cannot reveal hidden structure. They can, and RG types such structure as *hidden*, i.e. removable.
- That no conceivable observer outside physics could know everything. Such an observer is not a physical hypothesis.
- That everything beyond today's reach is permanently inaccessible. Only conditions (a)–(f) are claimed, each checkable.
- That the frameworks used are new. They are imported and cited. The contribution is their joint typing and the converted burden of proof.

18 Witnesses and limits

- `rg_window_witnesses.py` runs W1–W7, deterministically and with no random sampling. Reruns are byte-identical.
  - W1: the exact two-point instance.
  - W2: a symbolic twirl.
  - W3: pigeonhole and Fano.
  - W4: the event horizon by quadrature.
  - W5: exact orientation holonomy.
  - W6: exact boundary eigenvalues.
  - W7: 400-digit chaos.
- S1–S12 are taken from the Engagement Case battery.

- *Limits.* The horizon corollary is conditional on flat ΛCDM with constant Λ and $H_0 = 70$. The self-measurement and joint-measurability results are imported. The capacity condition uses declared finite memory, and the Bekenstein bound is cited only conditionally.

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